

Clinical Features and Initial Management Outcomes of Uveal Melanomas in a Single Tertiary Center in Egypt

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Keywords

Uveal melanoma · Ruthenium-106 brachytherapy · Gamma knife radiosurgery · Enucleation · Management outcomes · Egypt

Abstract

Introduction: We aimed to report the demographic data, clinical features, and management outcomes of patients with uveal melanoma (UM) in a single tertiary center in Egypt. **Methods:** This is a single-center retrospective case series. Patients with UM who were managed at Cairo University Ocular Oncology Service between January 2019 and December 2023 were included. Records were analyzed for patients' demographics, clinical features and different management options. Outcomes included changes in best corrected visual acuity and tumor dimensions (thickness and largest base diameter), perioperative complications, globe salvage, and recurrence rates. **Results:** A total of 93 eyes of 93 patients with UM were treated in our service over a period of 5 years. The mean age of patients was 52.18 ± 13.30 years, and 91.4% of tumors were choroidal or ciliochoroidal. 52 eyes (55.91%) were managed with ruthenium-106 episcleral brachytherapy, of which two eyes required additional bra-

chytherapy, 8 eyes ended with secondary enucleation, and the globe was salvaged in 84.62% of eyes. 8 eyes (8.60%) were managed by gamma knife radiosurgery with or without endoresection. One eye with an iridociliary tumor was managed with partial lamellar sclerouvectomy, and the remaining 32 eyes (34.41%) were primarily enucleated. **Conclusion:** Timely referral channels and accurate assessment of cases with UM are invaluable for achievement of higher success rates in tumor regression, as well as globe and/or vision salvage in eyes managed with brachytherapy. Patients with more advanced tumors should be counseled on the importance of more invasive options such as enucleation in achieving lower metastasis and mortality rates.

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Introduction

Uveal melanoma (UM) is the most common primary intraocular malignancy in adults with an age-adjusted incidence of 4.3 per million [1]. The choroid is involved in 85–90% of cases with UM, while the iris and ciliary body are involved in only 9–15%. Iris melanomas are usually discovered at an early phase and have a favorable

outcome; in contrast, ciliary body melanomas usually have a poor prognosis [2, 3].

With an incidence of 6–67% based on tumor size, the potential for systemic metastasis in UM has made primary treatment failure particularly dangerous and potentially lethal. It has been reported that patients with metastasis had a 15% chance of surviving 1 year and a median survival period of 4–15 months [4, 5].

For decades, primary enucleation remained the only viable option for management of UM. Despite earlier attempts, it was not until the Collaborative Ocular Melanoma Study (COMS) that brachytherapy gained wider acceptance as a potentially effective globe-salvaging tool in treating UM [6]. Nowadays, UM treatment options include brachytherapy, transpupillary thermotherapy (TTT), gamma knife radiosurgery (GKRS), and local tumor resection, either solely or in combination [7].

Available literature about the clinical features and management outcomes of UM in the Middle East are limited to a few retrospective series and case reports [8–10]. In this article, we aimed to highlight the clinical features and management outcomes of UM in a single tertiary center in Egypt.

Methods

This is a single-center retrospective review of the medical records of patients with UM managed at the Ocular Oncology Service, Ophthalmology department, Faculty of Medicine, Cairo University, in the period between January 2019 and December 2023. The study was compliant with the tenets of Helsinki Declaration and was approved by the Research Ethics Committee of Faculty of Medicine, Cairo University. Informed consent was obtained from all study subjects prior to any intervention.

Only patients with UM managed at our facility were included in the study. UM was diagnosed based on findings of clinical examination by binocular indirect ophthalmoscopy (by three senior ophthalmologists with experience in ocular oncology), in addition to the presence of characteristic echographic features of UM (dome or mushroom-shaped choroidal mass with homogeneous low to medium internal reflectivity, decay of echoes, and choroidal excavation).

Records of patients with inadequate data or with follow-up duration less than 6 months were excluded from analysis. Demographic data of patients including age, gender, and residence were recorded. At the initial visit and in subsequent follow-up visits, best corrected visual acuity (BCVA), intraocular pressure, and anterior segment were assessed, and dilated fundus examination was done in all patients.

All patients also had preoperative and postoperative colored fundus photography. Ultrasonography (US) was done at each visit to assess dimensions of the tumor and monitor change in echogenicity after treatment. Ultrasound biomicroscopy (UBM) was performed in cases with iris and ciliary body involvement.

Data about laterality, tumor type, location, macular involvement, proximity to optic disc, and associated features were recorded. The American Joint Committee on Cancer (AJCC) classification 8th edition was used to classify tumors based on largest base diameter (LBD) and height.

Patients with tumor thickness below 7 mm were managed by episcleral brachytherapy. Ruthenium-106 ophthalmic applicator (Eckert and Ziegler BEBIG GmbH, Berlin, Germany) was applied to the sclera overlying the tumor under general anesthesia. The tumor location was identified using transillumination or indirect binocular ophthalmoscopy and indentation. Intraoperative US was deployed to confirm proper plaque placement. Patients were kept hospitalized in a special isolated ward until the plaque was removed under general or local anesthesia after delivering the predetermined target dose of 120 Gray (Gy) to tumor apex.

In some patients with slightly thicker tumors (7–7.5 mm) and those adjacent to the optic nerve head margin, adjuvant TTT was applied on the surface or to the area adjacent to optic nerve. Patients with large tumors beyond the brachytherapy radiation range were candidates for primary enucleation, unless they opted for the alternative option of GKRS, which may be combined with endoresection.

All patients were followed up monthly in the first 6 months after intervention, then every 3 months for 2 years, then biannually. Systemic workup to detect metastasis including abdominal US and/or PET-CT scan was done in all patients on initial presentation and every 6 months.

Outcome measures included for all UM cases were metastasis, overall survival rate, and tumor-related mortality. Outcome measures for eyes receiving brachytherapy were globe salvage, changes in tumor dimensions, BCVA at last visit compared to baseline, nonresponse (defined as progressive growth by 15% on two consecutive monthly visits in the first 6 months after treatment), recurrences (defined as growth of tumor by more than 1 mm after an initial response to treatment), and treatment-related complications.

Statistical analysis was performed using Excel (Microsoft, Redmond, WA, USA). Continuous variables are presented as mean $\pm$ SD, while categorical variables are presented as frequency and percentage. Paired *t*-test was used to compare between baseline and last follow-up data. A two-sided *p* value <0.05 was considered to be statistically significant.

Results

In the period from January 2019 to December 2023, 115 patients were referred to the Ocular Oncology Service with a presumed diagnosis of UM, of which 16 (13.9%) were excluded due to a false referring diagnosis (online suppl. Table 1; for all online suppl. material, see <https://doi.org/10.1159/000540844>). Another 6 patients were excluded as they did not complete a minimum of 6-month follow-up.

Records of the remaining 93 patients with UM were included for analysis. 47 (50.54%) were females and 46 (49.46%) were males with no gender predilection. The

Table 1. Demographic data and clinical features of study subjects

Age, years (mean±SD)	52.18±13.30	Macular involvement ^a , <i>n</i> (%)	
Sex, <i>n</i> (%)		≤3 mm	26 (27.96)
Male	46 (49.46)	Distance from optic disc, <i>n</i> (%)	
Female	47 (50.54)	≤3 mm	42 (45.16)
Region, <i>n</i> (%)		Pigmentation, <i>n</i> (%)	
Greater Cairo region	26 (27.96)	Melanotic	90 (96.77)
Upper Egypt	26 (27.96)	Amelanotic	3 (3.23)
Delta region	38 (40.86)	Associated retinal detachment, <i>n</i> (%)	
Canal region	3 (3.22)	Yes	66 (70.97)
Baseline BCVA (mean±SD)	0.08±0.18	Baseline LBD, mm (mean±SD)	10.98±3.67
<0.1, <i>n</i> (%)	70 (75.27)	Baseline height, mm (mean±SD)	7.68±3.61
0.1–0.5, <i>n</i> (%)	19 (20.43)	≤7 mm, <i>n</i> (%)	36 (38.71)
>0.5, <i>n</i> (%)	4 (4.30)	>7 mm, <i>n</i> (%)	57 (61.29)
Involved eye, <i>n</i> (%)		AJCC classification, <i>n</i> (%)	
Right	51 (54.84)	T1	21 (22.58)
Left	42 (45.16)	T2	33 (35.48)
Site of tumor, <i>n</i> (%)		T3	29 (31.18)
Choroidal	72 (77.42)	T4	10 (10.76)
Ciliochoroidal	13 (13.98)	Primary intervention, <i>n</i> (%)	
Ciliary body	2 (2.15)	Brachytherapy	52 (55.91)
Iridociliary	6 (6.45)	Gamma knife radiosurgery	8 (8.60)
Location of tumor, <i>n</i> (%)		Primary enucleation	32 (34.41)
Superior	9 (9.68)	Partial lamellar sclerouvectomy	1 (1.08)
Inferior	10 (10.75)	Globe salvage, <i>n</i> (%)	
Nasal	15 (16.13)	Yes	52 (55.91)
Temporal	23 (24.73)	No	41 (44.09)
Superonasal	5 (5.38)		
Superotemporal	14 (15.05)		
Inferonasal	9 (9.68)		
Inferotemporal	8 (8.60)		

BCVA, best corrected visual acuity; LBD, largest base diameter. ^aTumor margin within 3 mm from the center of the fovea.

mean age was 52.18 ± 13.30 (range 24–74) years. Although our service was centered in Cairo, only 26 (27.96%) of the patients were residents of the catchment area, while the remaining patients (72.04%) were referred from remote governorates.

Mean BCVA in the involved eye was 0.08 ± 0.18 with 75.27% of eyes having BCVA less than 0.1. The right eye was involved in 51 patients (54.84%), and the choroid was involved in 85 patients (91.4%). The macula was involved in 26 patients (27.96%), and 45.16% of the lesions were within two disc diameters (<3 mm) from the optic disc (Table 1).

52 eyes (55.91%) were managed with brachytherapy using ruthenium plaques. The CCB plaque (circular with diameter 20.2 mm) was used in 25 eyes (48.08%), while the COB (notched with diameter 19.8 mm) and CIB (semilunar with diameter 20.2 mm) plaques were used for peripapillary tumors (26 eyes) and ciliary body tumors (1 eye), respectively. An apex dose of 120 Gy was delivered

to all patients with a mean treatment duration of 161 ± 96.69 (range 48–480) hours. Adjuvant TTT was applied at time of plaque insertion in 4 eyes.

After a mean follow-up duration of 17.54 ± 11.88 (range 6–50) months, the mean tumor height in the brachytherapy group decreased significantly from 5.14 ± 1.75 mm at baseline to 4.20 ± 1.65 mm at last visit (*p* value <0.001). Mean LBD also decreased significantly from 9.38 ± 2.99 at baseline to 7.77 ± 3.27 mm at last visit (*p* value <0.001). However, BCVA shows slight improvement from 0.11 ± 0.19 to 0.16 ± 0.24 (*p* value = 0.09) with only 36.54% of eyes showing one or more line improvement (Table 2).

Of the 52 eyes treated by brachytherapy, 5 eyes (9.62%), after an initial response, showed recurrence after 13.20 ± 6.57 months. Two of these eyes received a second episcleral brachytherapy and showed a good response with no evidence of recurrence throughout the follow-up

Table 2. Management outcomes in the brachytherapy group

Type of plaque, <i>n</i> (%)	
COB	26 (50)
CCB	25 (48.08)
CIB	1 (1.92)
Duration of application, h (mean±SD)	161.63±96.69
Mean follow-up duration, months (mean±SD)	17.54±11.88
BCVA (mean±SD)	
Baseline	0.11±0.19
Last follow-up	0.16±0.24
Change in BCVA, <i>n</i> (%)	
Improved	19 (36.54)
Stable	15 (28.84)
Worsened	18 (34.62)
LBD, mm (mean±SD)	
Baseline	9.38±2.99
Last follow-up	7.77±3.27
Height, mm (mean±SD)	
Baseline	5.14±1.75
Last follow-up	4.20±1.65
Complications, <i>n</i> (%)	
Retinal detachment	8 (15.38)
Neovascular glaucoma	7 (13.46)
Cataract	5 (9.62)
Radiation retinopathy	3 (5.77)
Toxic tumor syndrome	2 (3.85)
Radiation optic neuropathy	2 (3.85)
Scleral thinning and extrascleral extension	1 (1.92)
Endophthalmitis	1 (1.92)
Treatment failure	
Nonresponse	2 (3.85)
Recurrence	5 (9.62)
Time to recurrence, months (mean±SD)	13.20±6.57
Globe salvage rate, <i>n</i> (%)	
At 1 year	49 (94.23)
At 2 years	46 (88.46)
At 3 years	44 (84.62)
BCVA, best corrected visual acuity; LBD, largest base diameter.	

duration. The other three eyes opted for secondary enucleation.

Two more eyes (3.85%) showed nonresponse after brachytherapy and were also treated by secondary enucleation due to the peripapillary location of the tumor. Another two eyes, despite good tumor response to brachytherapy with reduction in tumor size, suffered from neovascular glaucoma and intolerable pain which necessitated enucleation.

One eye developed rhegmatogenous retinal detachment and endophthalmitis necessitating pars plana vit-

rectomy and eventually ended in a modified lid-sparing exenteration due to extensive extrascleral extension and invasive nature of the tumor. Of the 8 enucleated/exenterated eyes, histopathology showed epithelioid type in 4 eyes, spindle type in 1 eye, and necrotic tissue in 3 eyes. Globe salvage was achieved in 84.62% of patients with cumulative secondary enucleation rates of 5.77%, 11.54%, and 15.38% at 12, 24, and 36 months after treatment (online suppl. Table 2).

Thirty-two eyes (34.41%) were primarily enucleated due to the large baseline tumor dimensions incompatible

Table 3. Management outcomes in the primary enucleation group

AJCC T stage, <i>n</i> (%)	
T2	3 (9.38)
T3	21 (65.62)
T4	8 (25)
Baseline LBD, mm (mean±SD)	13.59±3.00
Baseline height, mm (mean±SD)	11.51±2.53
Histopathological type, <i>n</i> (%)	
Spindle cell	15 (46.86)
Epithelioid	5 (15.63)
Mixed	6 (18.75)
Melanocytoma	1 (3.13)
Unavailable	5 (15.63)
Extrascleral extension, <i>n</i> (%)	
Yes	2 (6.25)
No	30 (93.75)

with brachytherapy. One eye showed radiographic evidence of extrascleral extension and was managed by exenteration. Histopathological assessment of the enucleated eyes showed extrascleral extension in one more eye with optic nerve involvement, this eye was therefore treated with external beam radiotherapy, and follow-up PET-CT and abdominal US showed no evidence of local recurrence or distant metastasis (Table 3).

Eight patients (8.6%) with large tumors refused primary enucleation and hence were managed by GKRS, three of which had another surgery for endoresection within 7–10 days. Despite local tumor control in 87.5% of eyes, BCVA declined in 75% of eyes, and one eye showed recurrence after 6 months, making secondary enucleation the only viable option left for management (Table 4).

One patient with iridociliary melanoma was managed by exoresection through partial lamellar sclerouvectomy followed by application of ruthenium plaque to deliver a dose of 120 Gy to a height of 2 mm of scleral bed to reduce recurrence. The globe was salvaged with a final BCVA of 0.1. Despite the baseline and the 6 monthly metastatic workup in the form of abdominal US and/or PET-CT scan, distant metastasis was not detected in any of the patients in the current series.

Discussion

In the current study, we described the clinical features and management outcomes in a series of 93 Egyptian patients with UM who were treated at the

Table 4. Management outcomes in the gamma knife radio-surgery group

Gamma knife, <i>n</i> (%)	
Alone	5 (62.5)
With endoresection and plaque	3 (37.5)
Mean follow-up duration, months (mean±SD)	12±8.49
BCVA (mean±SD)	
Baseline	0.17±0.31
Last follow-up	0.08±0.15
Change in BCVA, <i>n</i> (%)	
Improved	2 (25)
Stable	0
Worsened	6 (75)
LBD, mm (mean±SD)	
Baseline	12.85±5.13
Last follow-up	12.30±4.83
Height, mm (mean±SD)	
Baseline	9.63±1.76
Last follow-up	9.89±2.12
Complications, <i>n</i> (%)	
Toxic tumor syndrome	3 (37.5)
Radiation retinopathy	2 (25)
Retinal detachment	2 (25)
Neovascular glaucoma	1 (12.5)
Radiation optic neuropathy	1 (12.5)
Recurrence	1 (12.5)
Globe salvage, <i>n</i> (%)	
Yes	7 (87.5)
No: secondary enucleation	1 (12.5)

BCVA, best corrected visual acuity; LBD, largest base diameter.

ocular oncology service in a single tertiary center over a period of 5 years. Even with scarce statistics about UM in the Middle East and most of the developing countries, there is an impression that incidence of UM in the Middle East is much lower than in the USA and Europe. Such wide variation may be attributed to the higher incidence of known risk factors as fair skin and light-colored eyes in the USA and Europe [9, 11].

The mean age in the current cohort is 52.18 ± 13.30 with no gender predilection which is consistent with international literature on choroidal melanoma [4, 12]. On the other hand, 75.27% of patients had a BCVA less than 0.1 on initial presentation, which is much worse than BCVA of patients studied in COMS where only 10% of patients had BCVA less than 0.1 [6], a finding which reflects the limited access of population to specialized healthcare. In the current cohort, only 27.96% of patients were referred from the catchment area served by our

tertiary center, while the majority of patients travel from remote regions, an observation which explains the delayed presentation and the poor baseline BCVA in the current series.

The lack of specialized ocular oncology services except in Cairo has many drawbacks on management outcomes, and a considerable proportion of patients presented with large-sized tumors beyond the capability of globe-salvaging treatment options. In addition, compliance of patients from remote areas to regular follow-up visits is compromised.

Ruthenium-106 brachytherapy was effective in local tumor control in our cohort with significant reduction in tumor size and globe salvage in 88.46 and 84.62% of eyes at 24 and 36 months after treatment. Similarly, Yaronov et al. [13] reported globe retention rates of 90% and 87% at 24 and 36 months.

Secondary enucleation was resorted to in 5.77%, 11.54%, and 15.38% of eyes at 12, 24, and 36 months after treatment. Bergman et al. [14] reported a 5-year cumulative incidence of enucleations of 16.8% with 75.5% of secondary enucleations performed in the first 3 years. Lower rates were, however, reported in COMS, and 5.2%, 7.7%, and 12.5% of treated patients underwent enucleation during the first 2, 3, and 5 years after treatment [6]. In a systematic review of 103 studies, the median 5-year local tumor control was lower with ruthenium (86%) compared to iodine (93%), mostly due to the suboptimal use of ruthenium, affecting target dose distribution [15].

A very wide range of doses to tumor apex was prescribed in literature (60–160 Gy) [15]. Although a dose of 85–100 Gy was widely accepted as a standard dose for iodine plaques, this may not be applicable for ruthenium [6, 16]. Ruthenium, being a beta emitter, has a very rapid dose falloff, resulting in significantly lower safety margins especially with tumors ≥ 5 mm thickness. To overcome this disadvantage, higher apex doses of 120–160 Gy were suggested to maximize the curative potential of ruthenium [17–19]. Increasing the apex dose from 100 Gy to 120 Gy was associated with increased globe salvage rates from 88 to 93% [17]. In another study using ruthenium plaques, a 5-year local control rate of 97.7% was achieved with 96% of patients receiving more than 100 Gy while 73% of patients received more than 125 Gy, where the high apex dose reduced the chances of local recurrence from any radioresistant tumor [20].

The effectiveness of combined use of brachytherapy and TTT has been demonstrated in several studies. Kreusel et al. demonstrated that ruthenium-106 bra-

chytherapy combined with TTT can successfully lead to local tumor control in tumors up to 8 mm in thickness. Tumor necrosis caused by TTT is induced to a depth of 3 mm from the apex [12, 13, 21].

With the unavailability of proton beam radiotherapy in Egypt, GKRS may represent a viable alternative option for globe salvage in patients with large tumors beyond the capacity of ruthenium brachytherapy. In the current cohort, local tumor control and globe salvage were achieved in 87.5% with recurrence in only 12.5% of cases treated by gamma knife. This is in line with other studies showing local control rates around 90% [22–25]. On the other hand, GKRS is associated with poor visual outcome and higher incidence of complications compared to brachytherapy [25–27]. In the current study, 75% of patients experienced deterioration in their BCVA.

However, vision deterioration after GKRS is only partly explained by radiation complications, and the tumor itself has a significant impact, due to macular involvement and associated vitreous hemorrhage or retinal detachment. Release of cytokines from irradiated ischemic tumor, known as toxic tumor syndrome, also contributes to poor postoperative BCVA, and such complications may be reduced by tumor endoresection within 7–10 days after gamma knife treatment [28].

Interestingly, in the current study, none of the patients showed any evidence of metastasis, and this may be due to the relatively short mean duration of follow-up in the study. Based on COMS, the rate of metastasis increases 10 years after initial diagnosis [29]. However, lower rates of metastasis have been reported in some middle eastern countries such as KSA (5%), Iran (6.8%), and Jordan (13%) compared to the USA and Europe which may shed light on a possible role of ethnicity in survival from UM [8, 9, 30]. Future multicenter longitudinal studies with molecular genetic analysis are needed to evaluate metastasis and metastasis-related mortality in our population.

In addition to the relatively short follow-up duration, the retrospective nature of the current study is another limitation. Standardized clinical assessment and systemic surveillance protocols applied at our center minimized bias related to retrospective study design to some extent.

Finally, we conclude that timely referral channels and accurate assessment of cases with UM are invaluable for achievement of higher success rates in tumor regression, as well as globe and/or vision salvage in eyes managed with brachytherapy. Patients with

more advanced tumors should be promptly counseled on the importance of more invasive options such as enucleation in achieving lower metastasis and mortality rates.

Statement of Ethics

All procedures performed were in accordance with the ethical standards of the Research Ethics Committee of Faculty of Medicine, Cairo University (N-5-2024). Patient consent was not required in accordance with the Local Research Ethics Committee guidelines for retrospective chart reviews.

Conflict of Interest Statement

All authors certify that they have no conflicts of interest in the subject matter or materials discussed in this manuscript.

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Author Contributions

I.Y.S., H.H., A.M.K., M.-S.H.E.-A., M.A.E.-H., T.A.M., D.H.H., S.H.S., A.M.N., A.E.F., Y.M., S.F.A.-E., L.E.Q., and A.M.A. contributed to the clinical, ancillary, and surgical care of the patients, as well as the retrospective data acquisition and analysis. I.Y.S. drafted the manuscript. T.A.M., A.M.A., and H.H. revised the manuscript. All authors approved the final version.

Data Availability Statement

The data that support the findings of this study are not publicly available due to privacy reasons but are available from the corresponding author upon reasonable request.

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